

Lab 4: Database Normalization and Implementation (3NF)

Tutoring Center System

Name: Eli Michel

Course: CS420

Part 1: Analyze the Current Design

1. Primary Key

The primary key of the original unnormalized table TUTORING_RECORD is SessionID because each tutoring session is uniquely identified by SessionID.

2. Functional Dependencies

- SessionID -> SessionDate, SessionTime, StudentID, TutorID, CourseID, RoomID, SessionType, HourlyRate, DurationMinutes
- StudentID -> StudentName, StudentEmail, Major
- TutorID -> TutorName, TutorEmail
- CourseID -> CourseTitle, Department
- RoomID -> RoomBuilding, RoomCapacity
- SessionID -> StudentName, StudentEmail, Major, TutorName, TutorEmail, CourseTitle, Department, RoomBuilding, RoomCapacity

3. Data Anomalies

Anomaly Type	Example
Insertion anomaly	A new tutor cannot be added unless the tutor has already conducted a tutoring session, because tutor information is stored only inside the tutoring record table.
Update anomaly	If a student changes their email address, the same email may need to be updated in many session rows. Missing one row would cause inconsistent student information.
Deletion anomaly	If the only session for a course is deleted, the system may also lose the course title and department information, even though the course itself still exists.

Part 2: Normalize to 3NF

The original table contains repeated student, tutor, course, and room data. To place the design into Third Normal Form, each real-world entity is moved into its own table, and the Session table stores only session-specific facts plus foreign keys.

Table	Attributes	Primary Key	Foreign Keys
Student	StudentID, StudentName, StudentEmail, Major	StudentID	None
Tutor	TutorID, TutorName, TutorEmail	TutorID	None
Course	CourseID, CourseTitle, Department	CourseID	None
Room	RoomID, RoomBuilding, RoomCapacity	RoomID	None

TutoringSession	SessionID, SessionDate, SessionTime, StudentID, TutorID, CourseID, RoomID, SessionType, HourlyRate, DurationMinutes	SessionID	StudentID references Student; TutorID references Tutor; CourseID references Course; RoomID references Room
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Why This Design Is in 3NF

- Each table has a primary key that uniquely identifies each row.
- Non-key attributes depend on the whole key of their table.
- Student details depend only on StudentID, tutor details depend only on TutorID, course details depend only on CourseID, and room details depend only on RoomID.
- The session table does not store repeated names, emails, course titles, departments, buildings, or room capacities. It stores foreign keys that connect to the correct entity tables.

Sample Table Instances

StudentID	StudentName	StudentEmail	Major
1001	Maria Lopez	maria.lopez@university.edu	Computer Science
1002	James Carter	james.carter@university.edu	Biology
1003	Aisha Khan	aisha.khan@university.edu	Business

TutorID	TutorName	TutorEmail
2001	Daniel Kim	daniel.kim@university.edu
2002	Sofia Martinez	sofia.martinez@university.edu
2003	Ethan Johnson	ethan.johnson@university.edu

CourseID	CourseTitle	Department
CS101	Introduction to Programming	Computer Science
MATH151	Calculus I	Mathematics
BIO110	General Biology	Biology

RoomID	RoomBuilding	RoomCapacity
R101	Science Hall	24
R102	Library Learning Center	18
R201	Technology Building	30

TutoringSession sample records - schedule and relationships:

SessionID	SessionDate	SessionTime	StudentID	TutorID	CourseID	RoomID
3001	2026-05-04	10:00:00	1001	2001	CS101	R201
3002	2026-05-04	11:30:00	1002	2003	BIO110	R101
3003	2026-05-05	14:00:00	1003	2002	BUS210	R102

TutoringSession sample records - session details:

SessionID	SessionType	HourlyRate	DurationMinutes
3001	Individual	25.0	60
3002	Group	22.5	90
3003	Individual	24.0	45

Part 5: Required Queries

Query #	Purpose
1	Display all tutoring sessions with student name, tutor name, course title, and room ID.
2	List all sessions for a specific student.
3	List all sessions conducted by a specific tutor.
4	Count the number of sessions per course.

Conclusion

This normalized design removes repeated data from the original tutoring record table. By separating students, tutors, courses, rooms, and sessions into different tables, the database reduces redundancy, prevents common data anomalies, and supports reliable MySQL queries through primary key and foreign key relationships.